DATA ANALYTICS PROJECT

A/B TESTING & STATISTICAL ANALYSIS

Evaluating the Impact of Checkout Page Design on Purchase Conversion Rate

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A/B Testing and Statistical Analysis

- **A/B Testing:** A randomized method to compare versions and identify which performs better.
- **Control Group (A)**: Existing version.
- **Treatment Group (B**): New version.

- **Types:**

1. **Standard A/B Test: Compares two versions (A & B).**
2. **A/B/n Test**: Compares multiple versions (A, B, C).
3. **Multivariate Test (MVT)**: Tests multiple variables simultaneously.

A/B Testing Example:

- Group A: Existing blue checkout page – 96 purchases.
- Group B: New orange checkout page – 108 purchases.
- Result: Compare purchases to identify the better design.
- Conclusion: Check statistical significance before final decision.

- **A/B Testing Use Cases:**

1. **Product & Website:** CTA buttons, onboarding page designs.
2. **Marketing & Advertising**: Email subject lines, ad designs, landing pages.
3. **E-commerce & Retail**: Free shipping, discount offers.
4. **Subscription & SaaS:** Monthly/annual pricing, free trial duration.
5. **Mobile Apps & Gaming**: Tutorials, push notifications.
6. **Search & Recommendations**: Search rankings, personalized recommendations.

- **How Does A/B Testing Work?**

1. **Define Problem:** Identify the business goal.
2. **Create Hypothesis**: Predict what change will improve results.
3. **Randomly Assign Users**: Divide users into Group A and Group B.
4. **Collect & Analyze Data**: Compare results using statistical analysis.
5. **Make a Decision**: Launch, reject, or test further.

Experiment Terminology

1. **Control Group (A):** Existing version; baseline for comparison.
2. **Treatment Group (B**): New version being tested.
3. **Hypothesis**: A statement to be tested.

- Null (H_0): No difference between A and B.
- Alternative (H_1): Difference exists between A and B.

4. **MDE (Minimum Detectable Effect)**: Smallest improvement an experiment can reliably detect. Smaller effects generally need larger samples.
5. **Primary Metric**: Main metric to measure success (e.g., conversion rate).
6. **Guardrail Metrics**: Metrics that should not worsen (e.g., refund rate, page load time).
7. **P-value**: Measures how unusual the observed result is if H_0 is true.

- p-value ≤ 0.05 → Statistically significant.
- p-value > 0.05 → Not statistically significant.

8. **Statistical Significance**: Evidence that the observed difference is unlikely due to random chance alone.

9. **Significance Level (α):** Risk of Type I error (false positive), usually 0.05.
   - Confidence Level = 1 − α = 95%.
10. **Statistical Power:** Probability of detecting an actual effect of a specified size.
    - Formula: Power = 1 − β
    - 80% Power → β = 20% (Type II error).
    - Higher power generally requires a larger sample size.

- **<u>Other Important A/B Testing Terms</u>**

1. **Randomization**: Randomly assigning users to groups to avoid selection bias.
2. **SRM (Sample Ratio Mismatch):** Actual group split differs from expected split (e.g., 60/40 instead of 50/50).
3. **Peeking:** Repeatedly checking results and stopping when a favourable p-value appears.
4. **Standard Error**: Measures uncertainty in a sample statistic.
5. **Sampling**: Testing a small portion of the total audience instead of every user.

- **Errors in A/B Testing:**

1. **Type I Error (False Positive – α**
- Concluding that a change works when no real improvement exists.
- Example: Launching a new page that actually performs no better than the existing page
- α = 0.05 (5%) – Common significance level.
- Simple: Launching a change that has no real effect.

2. **Type II Error (False Negative – β)**
- Failing to detect a real improvement.
- Example: Missing a genuine increase in conversion due to a small sample size.
- β = Type II Error Probability.
- Simple: Missing a real improvement.

- **Connection Between Sample Size, MDE, Errors & Power**

1. **Define Experiment:** Set baseline conversion rate and MDE.
2. **Choose Statistical Settings**: Set significance level (α) and power (1 – β).

3. **Calculate Sample Size & Duration:** Determine required users per group and experiment duration.
4. **Run & Analyze**: Measure the effect and check statistical significance

- **Statistical Methods to Analyze Test Result**

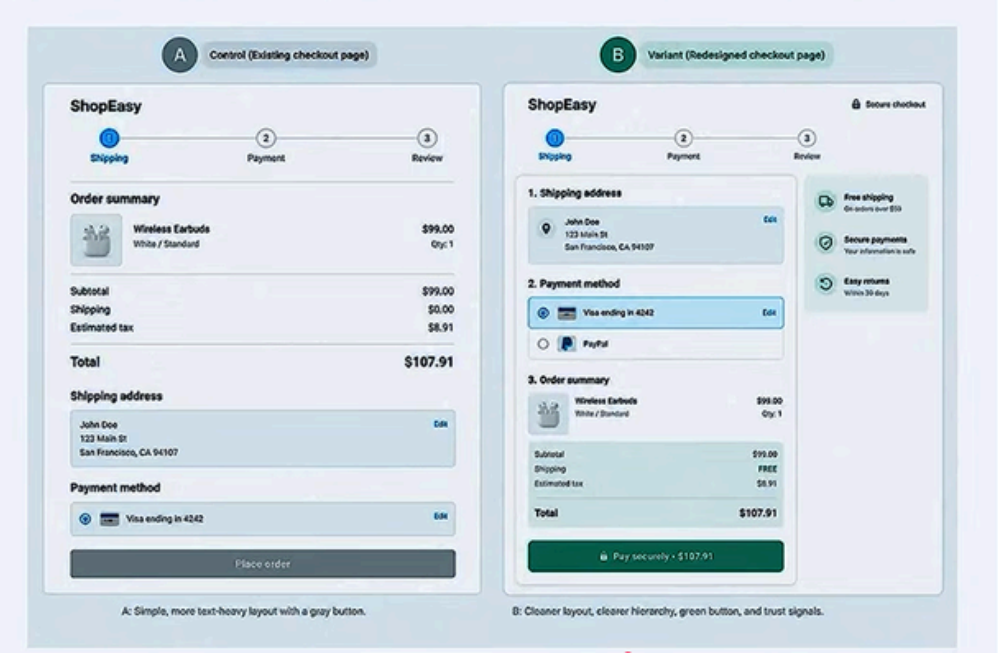
1. **Z-Test:** Compares proportions or percentages between two groups (large samples).
- Examples: Conversion Rate, CTR.
- Critical Z-value: ±1.96 (at 5% significance level, two-tailed).

2. **T-Test**: Compares averages (means) between two groups.
- Examples: Average Order Value (AOV), Revenue per User, Session Duration.

3. **Chi-Square Test**: Checks differences in categorical data or relationships between categories.
- Examples: User Preferences, Browser/Device Distribution.

Purchase Conversion Rate Experiment

- **Business scenario:** An e-commerce company wants to increase purchases by redesigning its checkout page.



- **Step 1: Define Objective & Hypothesis**
- Objective: Increase purchase conversion rate
- Control (A): Existing checkout page.
- Variant (B): Redesigned checkout page.
- Primary Metric: Purchase Conversion Rate = Purchases ÷ Eligible Users × 100.
- Guardrail Metrics: Average Order Value (AOV), Payment Failure Rate, Checkout Errors, Complaint Rate

- Null (H_0): No difference in conversion rates (A = B).
- Alternative (H_1): Conversion rates are different (A ≠ B).

- **Step 2: Set Experiment Parameters**
- Baseline Conversion Rate: 5%
- MDE: 1% (5% → 6%)
- Significance Level (α): 5%
- Statistical Power: 80%
- Allocation: 50% Group A / 50% Group B
- Sample Size: 8,160 users per group (16,320 total).
- Note: Sample size can be calculated using online tools or statistical formulas.

- **Step 3: Run the Experiment**
- Randomly assign users 50/50 to Group A and B.
- Keep each user's assigned version consistent.
- Run until required sample size and duration are met.
- Validate tracking, randomization and SRM.
- Avoid stopping early due to favourable results.

- **Step 4: Analyze Results**

Suppose the experiment completes with these results:

Metric	A: Existing	B: Redesigned
Users	10,000	10,000
Purchases	500	600
Conversion rate	5%	6%

- **Calculate the Impact**
- Absolute Lift: 6% − 5% = 1 percentage point.
- Relative Lift: (6% − 5%) ÷ 5% = 20%.
- Statistical Test: Two-Proportion Z-Test (Z = 3.1).
- P-value: 0.002.
- 95% Confidence Interval: +0.63 to +1.37 percentage points.
- Conclusion: CI is above zero and p-value < 0.05, so the result is statistically significant.

- **Step 5: Interpret & Recommend**
- Result: Group B performed better.
- Improvement: 20% relative increase in conversion rate.
- Decision: Statistically significant improvement.

- Check: Ensure guardrail metrics (AOV, payment success) are not negatively affected.
- Recommendation: Gradually roll out the new design and continue monitoring.

- **95% Confidence Interval (CI**
- Absolute Lift: 6% − 5% = 1 percentage point.
- Standard Error (SE): Measures uncertainty in the lift.
- Margin of Error: 1.96 × SE = 0.37%.
- 95% CI Formula: Absolute Lift ± Margin of Error.
- Lower Bound: 1% − 0.37% = +0.63%.
- Upper Bound: 1% + 0.37% = +1.37%.

- **Conclusion:**
- CI (+0.63% to +1.37%) is entirely above zero, indicating a positive improvement.

- **Acknowledgement**

The concepts and learning approach presented in this project are inspired by the A/B Testing educational tutorial by Rishabh Mishra on YouTube. The content has been organized and documented for learning purposes, with a focus on understanding statistical concepts, experiment design, and business interpretation.

Credit: Rishabh Mishra – YouTube Educational Content.